

Roll No.....

Odd End Semester Examination-2024

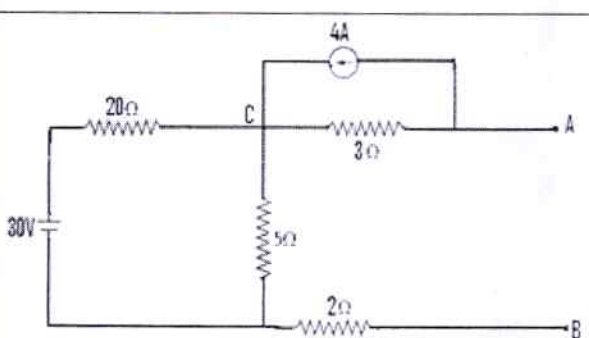
Btech (Biotech) Ist Semester

Basic Electrical and Electronics Engineering-EEC-101

Time: 3.00 Hrs Maximum Marks: 100

Note:

- (i) Answer all the questions by choosing any two of the sub questions.
- (ii) Each question carries 10 marks

Q1	(20 marks)	
(a)	Discuss the principle of operation of transformer. Also derive the emf equation of the transformer.	
(b)	Define the following terms: i. Magnetomotive force ii. Ampere turns iii. Reluctance iv. Permeance v. Reluctivity vi. Eddy current loss vii. Core loss	CO-01& CO-02
(c)	 Using Thevenin equivalent circuit across AB shown in fig	
Q2	(20 marks)	
(a)	Find average, RMS, Form factor and Peak factor of given waveform	CO-02& CO-03

(b)	Define the following terms - Real power - Apparent power - Reactive power - Power factor - Inductive reactance - Capacitive reactance Also specify the units of measurement for all the above mentioned terms.	
(c)	In the circuit shown in fig: find the current through 6 ohm resistor by using Nodal Analysis	
Q3	(20 marks)	
(a)	Discuss the transformer and its types in terms of voltage output. A single phase transformer has 600 turns on the primary and 80 turns on the secondary. If the primary is connected to a 3300 volt supply, find the secondary voltage. If the secondary current is 200A, what is the primary current?	
	Explain the R-L circuit with ac supply. Derive the expressions for current and instantaneous power. Draw the phasor diagram for the	

(b)	same.	CO-01 &CO-03
(c)	A ring has a mean diameter of 21 cm and a cross sectional area of 10 cm^2 . The ring is made up of semicircular sections of cast iron and cast steel, with each joint having a reluctance equal to an air gap of 0.2mm. Find the ampere turns required to produce a flux of $8 \times 10^{-4} \text{ Wb}$. The relative permeabilities of cast steel and cast iron are 800 and 166 respectively. Neglect leakage.	
Q4	(20 marks)	CO-04
(a)	(i) Distinguish between Avalanche breakdown and Zener breakdown. (ii) A Si diode has reverse saturation current of $2.0 \mu\text{A}$ at 300k. Find forward voltage to achieve a forward current of 10 mA. ($V_T = 26 \text{ mV}$)	
(b)	Discuss, with the help of a suitable diagram and waveforms the working of a center-tapped full wave rectifier and show that its ripple factor is 0.48.	
(c)	Differentiate between intrinsic and extrinsic semiconductor. Derive an expression for the conductivity of extrinsic and intrinsic semiconductors.	
Q5	(20 marks)	CO-05&06
(a)	With help of appropriate circuit diagrams, explain the input as well as output characteristics of Common Emitter Configuration of NPN transistor.	
(b)	Derive the relationship between: i. α , β and γ ii. I_{CEO} and I_{CBO} .	
(c)	Explain the working principle of N-P-N transistor with the help of a suitable diagram.	

